sorting are listed horizontally (from left to right). (C) A disconnected DAG; (D) A couple possible topological sorting for the graph in (C). (E) A chain graph: for this DAG, there is only one possible topological sorting.

DAGs and topological ordering have many fundamental applications in computer science, spanning from scheduling (at all levels) to resolving symbol dependencies in linkers.

14.3 Graph Traversal: BFS, DFS

To perform searches on a graph, as well as to apply many other algorithms, it is necessary to traverse the graph following its edges.

Like for trees, there are many possible ways to traverse a graph, depending on the order in which outgoing edges are traversed; in trees, however, it’s always clear where to start the traversal: from the root. By starting from the root, we are always sure that we can traverse a tree and reach all its nodes. In graphs, instead, there isn’t a special vertex alike to a tree’s root, and in general, depending on the vertex chosen as starting point, there might or might not be a sequence of edges that allows to visit all vertices.

In this section, we will focus on simple directed graphs; no other assumption will be made. We are not restricting ourselves to strongly connected graphs, and in general we don’t have further domain knowledge to choose the vertex from which we should start the search; consequently, we can’t guarantee that a single traversal will cover all the